

1 · DEFINE

HTN emergency = SBP >180 or DBP >120 + **acute end-organ damage**. The damage defines it, not the number. **Rate of rise > absolute value** — chronic HTN tolerates very high BP; a previously normotensive pt (eclampsia, acute GN) damages below 180/120.

2025 AHA/ACC: "hypertensive urgency" retired → **severe HTN without acute target-organ damage**. Mgmt unchanged: reinstitute/uptitrate oral meds, outpatient, close f/u. **No IV drip, no rapid drop.**

2 · END-ORGAN SCREEN

System	Emergencies / clues
Neuro	HTN encephalopathy (insidious HA/confusion/szr; PRES = parieto-occipital edema); ICH/SAH & ischemic CVA (focal deficits); papilledema
Cardiac	Acute LV failure + pulm edema; ACS; aortic dissection (tearing pain to back, inter-arm BP diff, ↓pulses, widened mediastinum)
Renal	Rapidly ↑Cr, ↓UOP, microscopic hematuria
Other	Eclampsia (>20 wk + szr); pheo crisis (episodic HA/sweats/palpitations/tachy/hyperthermia); sympathomimetics (+ tox); VEGF inhibitors; TMA

3 · TREATMENT GOALS

Goal = **prevent further damage, NOT normalize BP**. ICU + IV titratable gtt; remove exacerbators (pain, volume, drugs). Autoregulation has reset → fast drop = hypoperfusion.

Scenario	Target / pace
General	↓SBP ≤25% in 1st h → ~160/100 by 2–6 h → nl over 24–48 h
Dissection	HR ~60 w/ BB <u>first</u> , then SBP <120 ASAP
ICH / SAH	SBP <140 promptly
Eclampsia / pheo	SBP <140 in 1st h (+ MgSO4 for eclampsia)
Ischemic CVA	Permissive ≤220/120; if higher ↓~15%/24h; tPA cand <185/110 pre, <180/105 ×24h

Dissection: rate before pressure — vasodilator first triggers reflex tachycardia → ↑wall shear (dP/dt) → extends dissection.

4 · DRUGS BY MECHANISM

Agent	Class	Use for / caution
Nicardipine	Art. vasodil (CCB)	Stroke, pregnancy; broad
Hydralazine	Arterial	Pregnancy; reflex tachy (ACS), ↑ICP, unpredictable
Nitroprusside	Art + venous	Almost any (not preg); CN tox in AKI, ↑ICP
Fenoldopam	D1 agonist	Acute renal failure ; ↑ICP/IOP
Nitroglycerin	Venous (↓preload)	ACS, pulm edema ; weak antiHTN
Labetalol	α1/β	Eclampsia, ACS, dissection; not RV infarct / unknown EF
Esmolol	β1	Dissection (+ vasodil)
Phentolamine	α1	Catecholamine excess (pheo, sympathomimetic, clonidine w/d)

SCENARIO → AGENT (quick pick)

- Ischemic stroke / pregnancy: nicardipine
- ACS / pulmonary edema: nitroglycerin ± nitroprusside + diuresis
- Aortic dissection: esmolol/labetalol (HR 60) then nitroprusside
- Pheo / sympathomimetic: phentolamine (then maybe nicardipine/nitroprusside)
- Acute renal failure: fenoldopam or nicardipine

DO / AVOID

AVOID: β-blocker before EF known in acute LV failure (→ cardiogenic shock); β-blocker as initial Rx in pheo/ catecholamine excess (unopposed α); nitroprusside in AKI (cyanide); hydralazine in ACS (reflex tachy); rapid normalization in any emergency.

DO: confirm organ damage before calling it an emergency; lower gradually to target; treat the driver (pain, missed meds, tox, eclampsia); ICU + a-line for any gtt.